

Phase 5 – Project Development

Introduction

This phase describes the implementation of the Personalized Networking Assistant, including the project modules, technologies, APIs, and application workflow.

Project Modules

- 1. User Interface (Streamlit)
- 2. FastAPI Backend
- 3. Theme Extraction using DistilBERT
- 4. Conversation Starter Generation using GPT-2
- 5. Fact Verification using Wikipedia API
- 6. History & Feedback Management

Development Workflow

The user enters event details and interests. The backend extracts themes, generates conversation starters, validates them, optionally verifies facts, stores history, and displays the results through the Streamlit interface.

Key Features

- AI-generated conversation starters
- Theme extraction
- Wikipedia-backed fact verification
- Feedback collection
- Conversation history
- Modular architecture

Technology Stack

Programming Language: Python
Frontend: Streamlit
Backend: FastAPI
AI Models: DistilBERT, GPT-2
Database: Local JSON Storage
Testing: Pytest
Version Control: Git & GitHub

Folder Structure

backend/
frontend/
data/
tests/
requirements.txt
README.md

Module	Description
Frontend	Accepts user input and displays generated conversation starters
Backend	Handles API requests and coordinates services
Theme Extractor	Analyzes event descriptions using DistilBERT

Starter Generator	Generates personalized conversation starters using GPT-2
Fact Checker	Verifies claims using Wikipedia API
History Logger	Stores interaction history and user feedback